

CATEGORY		LOW-END INTEL MOTHERBOARDS						
FEATURES (20 %)	Brand	AOpen	ASRock	ASUS	Gigabyte	Gigabyte	HIS	HIS
	Model	MX4GVR-GN	P4I45GV	P4BGV-MX	GA-8I845GV	GA-8GEM667	M48-E1	M53-1A
	Chipset	Intel 845GL	Intel 845GV	Intel 845GV	Intel 845GV	845GE	Intel 845GLX	Intel 845GV
	Interface type	Socket 478	Socket 478	Socket 478	Socket 478	Socket 478	Socket 478	Socket 478
PERFORMANCE (80 %)	CPU speed (GHz) / FSB supported (MHz)	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533
	Memory speed (MHz)/ Memory slots (Nos)	DDR 333 / 2	DDR 333 / 2	DDR 333 / 2	DDR 266 / 2	DDR 333 / 2	DDR 266 / 2	DDR 266 / 2
	Form factor	Micro ATX	Micro ATX	Micro ATX	Micro ATX	Micro ATX	Micro ATX	Micro ATX
	AGP slot / Onboard AGP	X/✓	X/✓	X/✓	X/✓	✓/✓	X/✓	X/✓
	PCI Slots	3	2	3	3	3	2	2
	Onboard Sound	✓	✓	✓	✓	✓	✓	✓
	USB ports / Rear / Ports Via Cable	6/4/2	6/4/2	6/4/2	6/2/4	6/2/4	6/2/4	6/2/4
	Ethernet	✓	✓	✓	✓	✓	✓	X
	HyperThreading Support	✓	X	✓	✓	✓	✓	X
	Core voltage adjustment	✓	X	✓	✓	✓	X	X
OVERALL SCORE	Frequency adjustment	✓	X	✓	✓	✓	X	X
	Component layout on the motherboard (S05)*	3	3	2.5	2.5	3	3	3
	Package contents							
	Manual / CDs	✓/✓	✓/✓	✓/✓	✓/✓	✓/✓	✓/✓	✓/✓
	Necessary cables provided	✓	✓	✓	✓	✓	✓	✓
	Gaming Benchmarks: Serious Sam Second Encounter							
	Average fps	25.70	28.47	32.10	27.80	31.67	26.97	26.77
	Quake III Arena							
	Average fps	68.50	78.93	80.77	60.00	72.33	66.13	68.70
	Real World Benchmarks							
OVERALL SCORE	ZDbench Content Creation 2004	22.7	Failed	Failed	22.1	22.7	22.7	22.6
	ZDbench Business Winstone 2004	18.9	19.3	20	17.5	21.3	18.4	18.1
	Video Encoding (Time in secs)	103	95	89	106	89	101	100
	Synthetic Benchmarks							
	3D Mark 2001 SE (patched for onboard without AGP slot)	1,545	1,533	1,836	1,422	1,532	1,534	2,737
	PCMark 04							
	CPU index	3,564	4,320	4,079	3,589	3,627	3,568	3,569
	Memory index	2,397	2,508	2,279	1,661	3,178	2,363	2,411
	Graphics	483	539	617	438	498	475	489
	Hard disk index	3,321	3,300	3,384	3,298	3,865	3,329	3,351
OVERALL SCORE	Total Score	2,766	3,390	3,228	2,557	3,745	2,731	2,802
	Sisoft Sandra 2004 Professional: CPU sub-system							
	CPU Dhrystone/Whetstone	7,175 / 2,051	9,481 / 3,676	7,889 / 2,262	7,128 / 2,058	7,202 / 2,070	7,023 / 2,055	7,073 / 2,055
	Multimedia - CPU Integer / FPU SSE	17,240 / 20,424	23,570 / 32,950	19,042 / 22,519	17,297 / 20,396	17,373 / 20,586	17,244 / 20,409	17,240/20,421
	Memory sub-system							
	ALU / RAM	1,648	1,663	2,197	1,475	2,143	1,637	1,656
	FPU / RAM	1,651	1,661	2,198	1,474	2,145	1,637	1,657
	Disk drive							
	Drive Index MBps	24,588	26,500	26,535	26,775	26,543	26,340	27,654
	Performance	46.44	41.35	42.86	44.37	51.53	46.38	50.10
OVERALL SCORE	Features	12.80	9.80	9.50	11.50	13.80	9.80	7.80
	Grand Total (Performance + Features)	59.24	51.15	52.36	55.87	65.33	56.18	57.90
	VFM (Index)	12.34	17.05	13.25	16.43	13.33	18.12	17.54
	Price (Rs)	4,800	3,000	3,950	3,400	4,900	3,100	3,300
	Grade	B	B	B	B	B	B	B

*Scale of 5

Know the Plan

Many enthusiasts like to install a larger heatsink fan on their motherboards for assured cooling. In fact, some of them go to the extent of actually investing an extra thousand bucks for the sake of a cooling fan.

However, picture this: You get the coolest looking cooling-combo from abroad, and to your dismay find that there is not enough space around the CPU socket to plug the cooler. This is a common problem with the motherboard

layouts. Lesson learned: Always make that the CPU area is devoid of any obstructive capacitor so that you can use a larger heatsink fan combo. Some manufacturers resort to another implementation that places the heatsink perpendicular to the line of capacitors around the CPU area—a good design since the capacitors are on the sides and do not come in the way, when installing or removing the combo.

Another common problem is the placement of the AGP slot relative to the

memory slots. Many badly designed boards have the AGP-slot lock interfering with the memory-slot locks. This makes it impossible to remove the RAM modules without unplugging the display card first. Also, certain boards have a tiny capacitor near the AGP lock that is a constant source of irritation when unplugging high-end display cards such as the GeForce4 Ti.

If you use a standard-size ATX cabinet, check the position of the ATX power